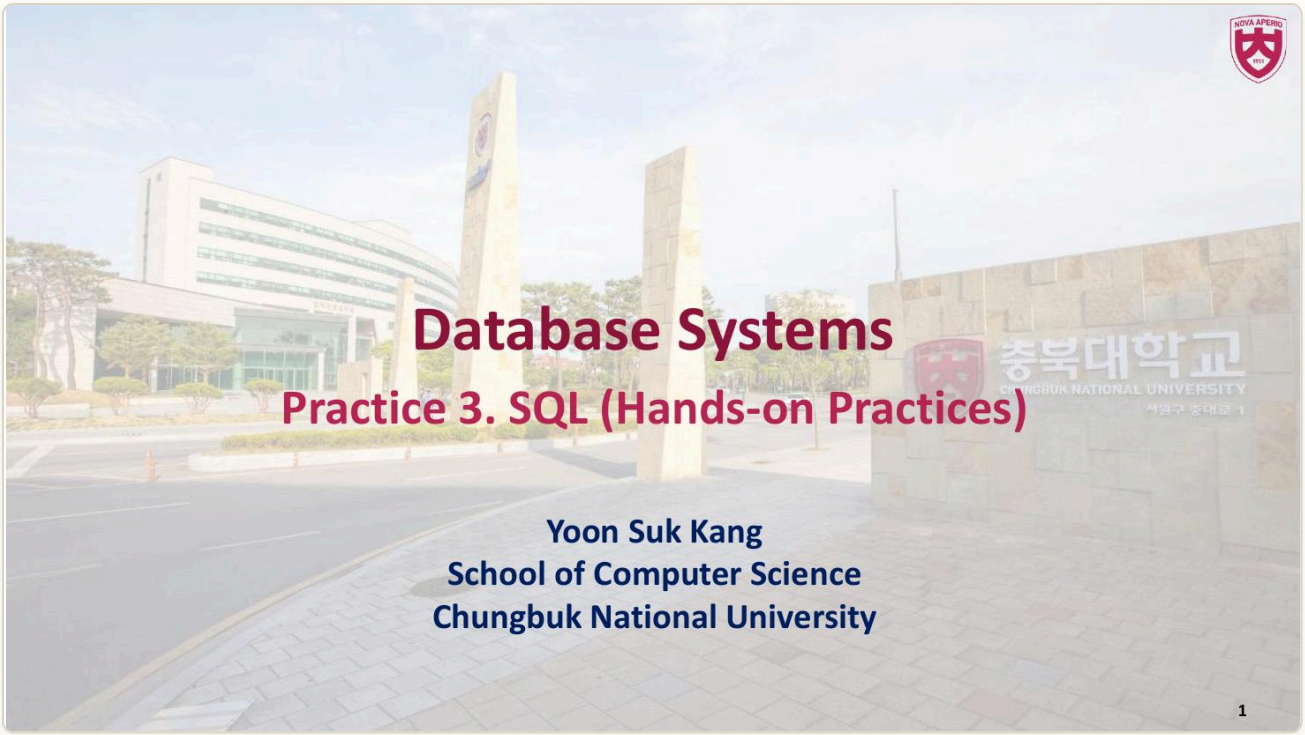
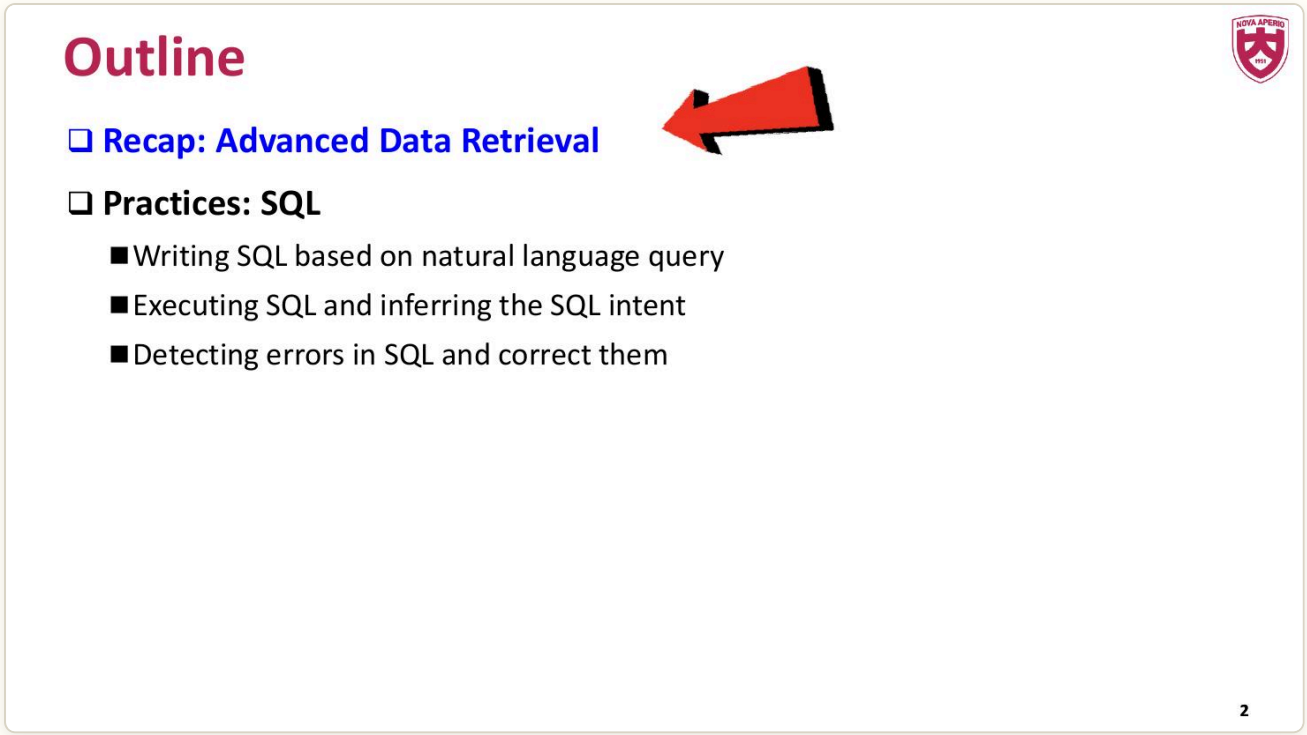


데이터베이스 시스템 — 실습 3: SQL

- Yoon Suk Kang · 충북대 소프트웨어학부

✍️ 메모

💡 이해 4강 고급 SQL(중첩 질의·조인·집계·그룹)을 손으로 써 보고, 실행 결과를 계산하고, 틀린 SQL을 고치는 실습



목차 (Outline)

- **복습: 고급 데이터 검색**
- **실습: SQL** — 자연어 질의를 SQL로 쓰기 / SQL을 실행해 의도 추론하기 / SQL의 오류를 찾아 고치기

✍ 메모

— 넘어가도 되는 쪽

Recap : Relational Data Model

❑ **Nested Query**
- SELECT-FROM-WHERE Blocks within a WHERE clause of another query

❑ **Joined Table**
- Inner and outer joins

❑ **Aggregate Functions**
- Summarize information from multiple tuples

❑ **Grouping**
- Partition **groups** of tuples in a relation

```
SELECT <attribute and function list>
FROM <table list>
[ WHERE <condition> ]
[ GROUP BY <grouping attribute(s)> ]
[ HAVING <group condition> ]
[ ORDER BY <attribute list> ];
```

FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY

Execution order

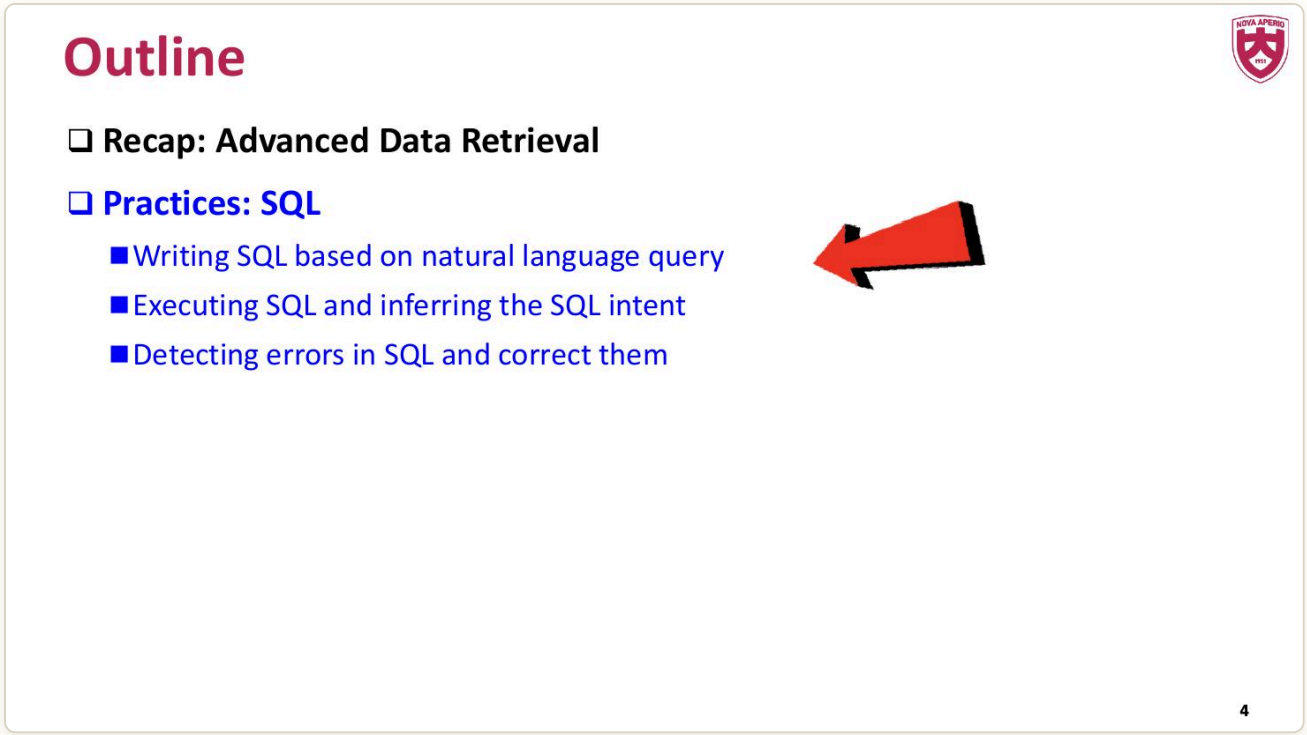
3

복습: 고급 데이터 검색

- **Nested Query** (중첩 질의): 다른 질의의 WHERE 안에 든 SELECT-FROM-WHERE 블록
- **Joined Table** (조인된 테이블): 내부 조인과 외부 조인
- **Aggregate Functions** (집계 함수): 여러 튜플의 정보를 요약
- **Grouping** (그룹화): 릴레이션의 튜플을 그룹으로 나눔

✍ 메모

★ 시험 아래 연습은 이 넷을 조합하는 문제 — 특히 NOT EXISTS, 서브쿼리 안 MAX/MIN/AVG



목차 (Outline)

- **복습: 고급 데이터 검색**
- **실습: SQL** — 자연어 질의를 SQL로 쓰기 / SQL을 실행해 의도 추론하기 / SQL의 오류를 찾아 고치기

✍ 메모

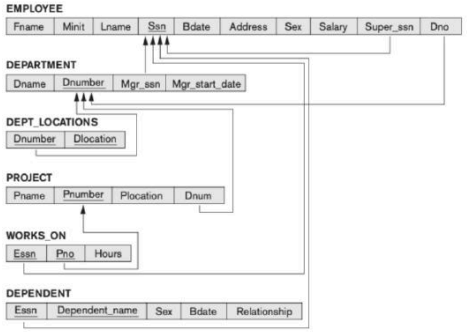
— 넘어가도 되는 쪽

Writing SQL based on natural language query

❑ Exercise 1

■ Specify the following queries on the COMPANY database using the concept of nested queries, grouping, aggregate functions, etc.

❑ Q1-1. Retrieve the names of all employees who work in the department that has the employee with the highest salary among all employees



5

연습 1 Q1-1

- COMPANY DB에서 중첩 질의·그룹·집계 등을 써서
- **Q1-1** 전체 직원 중 월급이 가장 높은 직원이 있는 부서에서 일하는 모든 직원의 이름

✍ 메모

✏ 예·풀이 SELECT Fname, Minit, Lname FROM EMPLOYEE WHERE Dno = (SELECT Dno FROM EMPLOYEE WHERE Salary = (SELECT MAX(Salary) FROM EMPLOYEE));

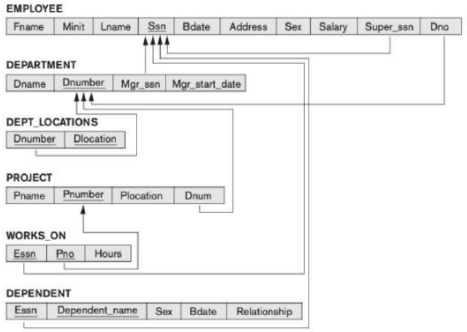
★ 시험 결과: 최고 55000 = Borg(Dno 1) → **James E Borg** 한 명

Writing SQL based on natural language query

❑ Exercise 1

■ Specify the following queries on the COMPANY database using the concept of nested queries, grouping, aggregate functions, etc.

❑ Q1-2. Retrieve the names of all employees whose supervisor's supervisor has '88866555' for Ssn.



6

연습 1 Q1-2

- **Q1-2** 상사의 상사의 Ssn이 '888665555'인 모든 직원의 이름

✍ 메모

✏ 예·풀이 SELECT Fname, Minit, Lname FROM EMPLOYEE WHERE Super_ssn IN (SELECT Ssn FROM EMPLOYEE WHERE Super_ssn = '888665555');

★ 시험 결과: **Smith, Narayan, English** (Wong의 부하) + **Zelaya, Jabbar** (Wallace의 부하)

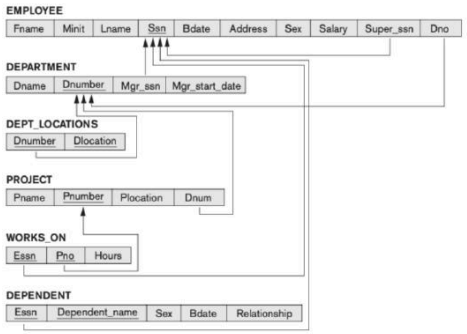
⚠ 주의 슬라이드의 '88866555'는 '888665555' 오타. 해답은 = 을 썼지만 서브쿼리가 2행이라 IN 이 안전

Writing SQL based on natural language query

❑ Exercise 1

■ Specify the following queries on the COMPANY database using the concept of nested queries, grouping, aggregate functions, etc.

❑ Q1-3. Retrieve the names of employees who make at least $10,000 more than the employee who is paid the least in the company



7

연습 1 Q1-3

- **Q1-3** 회사에서 가장 적게 받는 직원보다 $10,000 이상 더 받는 직원의 이름

✍ 메모

✏ 예·풀이 SELECT Fname, Minit, Lname FROM EMPLOYEE WHERE Salary - 10000 >= (SELECT MIN(Salary) FROM EMPLOYEE);

★ 시험 최저 25000 → 35000 이상: **Wong 40000, Wallace 43000, Narayan 38000, Borg 55000**

Writing SQL based on natural language query

❑ Exercise 2

■ Specify the following query on the UNIVERSITY database schema in SQL

❑ Q2-1. Retrieve the names and major department of all straight-A students who have a grade of A in any of their courses

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

연습 2 Q2-1 (UNIVERSITY)

- **Q2-1** 모든 과목에서 A를 받은(straight-A) 학생의 이름과 전공
- 스키마: STUDENT, COURSE, SECTION, GRADE_REPORT, PREREQUISITE

✍ 메모

✏ 예·풀이 SELECT S.Name, S.Major FROM STUDENT AS S WHERE NOT EXISTS (SELECT * FROM GRADE_REPORT AS G WHERE G.Student_number = S.Student_number AND G.Grade <> 'A');

★ 시험 '모두 A' = **A가 아닌 성적이 하나도 없다** → NOT EXISTS (... <> 'A'). 교안 DB에선 Brown에게 B가 있어 결과 없음

⚠ 주의 해답 슬라이드의 'NOT EXIST'는 NOT EXISTS 오타

Writing SQL based on natural language query

❑ Exercise 2

■ Specify the following query on the UNIVERSITY database schema in SQL

❑ Q2-2. Retrieve the names and major departments of all students who do not have a grade of A in any of their courses

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

연습 2 Q2-2

- **Q2-2** 어떤 과목에서도 A를 받지 못한 학생의 이름과 전공

✍ 메모

✏ 예·풀이 ... WHERE NOT EXISTS (SELECT * FROM GRADE_REPORT AS G WHERE G.Student_number = S.Student_number AND G.Grade = 'A');

★ 시험 Q2-1과 차이는 **<> 'A'** 대 **= 'A'** 하나. 결과: **Smith, CS** (B, C만 받음)

Inferring SQL Intent and Output

❑ **Exercise 3: Given an SQL query,**

■(1) write the result obtained after executing the query on the following DB state

```
SELECT P.Pname, P.Plocation
FROM PROJECT P
WHERE P.Dnum = ( SELECT D.Dnumber
                 FROM DEPARTMENT D
                 WHERE D.Dname = 'Administration' );
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 3 (1) 실행 결과 쓰기

- SELECT P.Pname, P.Plocation FROM PROJECT P WHERE P.Dnum = (SELECT D.Dnumber FROM DEPARTMENT D WHERE D.Dname = 'Administration');

✍ 메모

★ 시험 Administration = 4번 부서 → Computerization (Stafford), Newbenefits (Stafford)

Inferring SQL Intent and Output

❑ **Exercise 3: Given an SQL query,**

■ (2) infer the original natural language query (query intent)

```
SELECT P.Pname, P.Plocation
FROM PROJECT P
WHERE P.Dnum = ( SELECT D.Dnumber
                 FROM DEPARTMENT D
                 WHERE D.Dname = 'Administration' );
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 3 (2) 의도 추론

- 원래 자연어 질의(의도)를 추론하라

✍ 메모

★ 시험 의도: **'Administration' 부서가 담당하는 모든 프로젝트의 이름과 위치**를 검색

Inferring SQL Intent and Output

❑ **Exercise 4: Given an SQL query,**

■(1) write the result obtained after executing the query on the following DB state

```
SELECT E.Fname, E.Lname
FROM EMPLOYEE AS E
WHERE  NOT EXISTS (
            ( SELECT W.Pno
              FROM WORKS_ON AS W
              WHERE W.Essn = E.Ssn)
            EXCEPT ( SELECT P.Pnumber
                     FROM PROJECT AS P
                     WHERE P.Dnum = E.Dno) );
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 4 (1) 실행 결과 쓰기

- SELECT E.Fname, E.Lname FROM EMPLOYEE AS E WHERE NOT EXISTS ((SELECT W.Pno FROM WORKS_ON AS W WHERE W.Essn = E.Ssn) EXCEPT (SELECT P.Pnumber FROM PROJECT AS P WHERE P.Dnum = E.Dno));

✍ 메모

★ 시험 결과: John Smith, Joyce English, Ramesh Narayan, James Borg, Ahmad Jabbar, Alicia Zelaya

💡 이해 빠지는 사람: Wong(5번)은 10·20번(4·1번 부서) 프로젝트에도, Wallace(4번)는 20번(1번 부서)에도 참여

Inferring SQL Intent and Output

❑ **Exercise 4: Given an SQL query,**

■(2) infer the original natural language query (query intent)

```
SELECT E.Fname, E.Lname
FROM EMPLOYEE AS E
WHERE  NOT EXISTS (
            ( SELECT W.Pno
              FROM WORKS_ON AS W
              WHERE W.Essn = E.Ssn)
            EXCEPT  ( SELECT P.Pnumber
                      FROM PROJECT AS P
                      WHERE P.Dnum = E.Dno) );
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 4 (2) 의도 추론

- 원래 자연어 질의(의도)를 추론하라

✍️ 메모

★ 시험 의도: 자신이 속한 부서에서 관리하는 프로젝트에만 참여하는 모든 직원의 이름과 성

💡 이해 (내 프로젝트) EXCEPT (내 부서 프로젝트)가 비었다 = 부분집합 — 나눗셈과 같은 NOT EXISTS 패턴

Inferring SQL Intent and Output

❑ **Exercise 5: Given an SQL query,**

■ (1) write the result obtained after executing the query on the following DB state

```
SELECT E.Fname, E.Lname, E.Salary
FROM EMPLOYEE AS E
WHERE E.Salary < (
   SELECT M.Salary
   FROM EMPLOYEE AS M
   WHERE M.Ssn = (
      SELECT D.Mgr_ssn
      FROM DEPARTMENT AS D
      WHERE D.Dnumber = E.Dno )
)
AND E.Salary > (
   SELECT AVG(E2.Salary)
   FROM EMPLOYEE AS E2
   WHERE E2.Dno = E.Dno)
ORDER BY E.Dno ASC, E.Salary DESC;
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 5 (1) 실행 결과 쓰기

- E.Salary < (부서 관리자의 월급) AND E.Salary > (같은 부서의 AVG(Salary)) ORDER BY E.Dno ASC, E.Salary DESC

✍ 메모

✏ 예·풀이 5번 부서 평균 (30000+40000+38000+25000)/4 = 33250, 관리자 Wong 40000 / 4번 평균 31000, 관리자 Wallace 43000

★ 시험 결과: **Ramesh Narayan 38000** 한 명

Inferring SQL Intent and Output

❑ **Exercise 5: Given an SQL query,**

■(2) infer the original natural language query (query intent)

```
SELECT E.Fname, E.Lname, E.Salary
FROM EMPLOYEE AS E
WHERE E.Salary < (
   SELECT M.Salary
   FROM EMPLOYEE AS M
   WHERE M.Ssn = (
      SELECT D.Mgr_ssn
      FROM DEPARTMENT AS D
      WHERE D.Dnumber = E.Dno )
)
AND E.Salary > (
   SELECT AVG(E2.Salary)
   FROM EMPLOYEE AS E2
   WHERE E2.Dno = E.Dno)
ORDER BY E.Dno ASC, E.Salary DESC;
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 5 (2) 의도 추론

- 원래 자연어 질의(의도)를 추론하라

✍ 메모

★ 시험 의도: 자신이 속한 부서의 평균 급여보다 높고 해당 부서 관리자의 급여보다 낮은 모든 직원의 이름·성·급여, 부서 번호 오름차순·급여 내림차순

Detecting Errors in SQL

❑ **Exercise 6: Given an SQL query,**

■(1) write the result obtained after executing the query on the following DB state

Q. For each project, retrieve the project number, project name, and the number of employees who work more than 20 hours.
Sort the results by the project number in ascending order

```
SELECT P.Pnumber, P.Pname, COUNT(*)
FROM PROJECT AS P JOIN WORKS_ON AS W
  ON P.Pnumber = W.Pno
WHERE W.Hours >= 20
GROUP BY P.Pnumber, P.Pname
ORDER BY Pnumber ASC;
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 6-1 (1) 실행 결과 쓰기

- Q. 각 프로젝트의 번호·이름·20시간 넘게 일하는 직원 수, 번호 오름차순
- SELECT P.Pnumber, P.Pname, COUNT(*) FROM PROJECT AS P JOIN WORKS_ON AS W ON P.Pnumber = W.Pno WHERE W.Hours >= 20 GROUP BY P.Pnumber, P.Pname ORDER BY Pnumber ASC;

✍️ 메모

★ 시험 결과: 1 ProductX 2 / 2 ProductY 1 / 3 ProductZ 1 / 10 Computerization 1 / 30 Newbenefits 2

Detecting Errors in SQL

❑ **Exercise 6: Given an SQL query,**

■(2) find errors and correct them

Q. For each project, retrieve the project number, project name, and the number of employees who work more than 20 hours.
Sort the results by the project number in ascending order

```
SELECT P.Pnumber, P.Pname, COUNT(*)
FROM PROJECT AS P JOIN WORKS_ON AS W
  ON P.Pnumber = W.Pno
WHERE W.Hours >= 20
GROUP BY P.Pnumber, P.Pname
ORDER BY Pnumber ASC;
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 6-1 (2) 오류 찾고 고치기

- 오류를 찾아 고쳐라

✍ 메모

⚠ 주의 오류: 20시간 이상인 직원이 0명인 **Reorganization(20)** 이 결과에서 빠진다 (내부 조인 + WHERE)

✏ 예·풀이 고침: FROM PROJECT AS P **LEFT JOIN** WORKS_ON AS W ON (P.Pnumber = W.Pno **AND W.Hours >= 20**) … **COUNT(W.Essn)** → Reorganization 0 추가

💡 이해 COUNT(*)면 NULL 행도 1로 센다 — 열 이름으로 세야 0

Detecting Errors in SQL

❑ **Exercise 6: Given an SQL query,**

■(1) write the result obtained after executing the query on the following DB state

Q. For every employee who works on at least one project and earns a salary higher than the average salary of the department the employee works for, retrieve the employee's name, department number, and salary

```
SELECT E.Fname, E.Lname, E.Dno, E.Salary
FROM EMPLOYEE AS E JOIN WORKS_ON AS W
   ON E.Ssn = W.Essn
WHERE E.Salary > (
   SELECT AVG(E2.Salary)
   FROM EMPLOYEE AS E2
   JOIN WORKS_ON AS W2
     ON E2.Ssn = W2.Essn
   WHERE E2.Dno = E.Dno )
ORDER BY Salary DESC;
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 6-2 (1) 실행 결과 쓰기

- Q. 프로젝트에 하나 이상 참여하고 자기 부서 평균 월급보다 많이 받는 직원의 이름·부서 번호·월급
- SELECT … FROM EMPLOYEE AS E JOIN WORKS_ON AS W ON E.Ssn = W.Essn WHERE E.Salary > (SELECT AVG(E2.Salary) FROM EMPLOYEE AS E2 JOIN WORKS_ON AS W2 ON E2.Ssn = W2.Essn WHERE E2.Dno = E.Dno) ORDER BY Salary DESC;

✍ 메모

★ 시험 결과: Wallace 43000 ×2, Wong 40000 ×4, Narayan 38000 (참여 프로젝트 수만큼 중복)

Detecting Errors in SQL

❑ **Exercise 6: Given an SQL query,**

■(2) find errors and correct them

Q. For every employee who works on at least one project and earns a salary higher than the average salary of the department the employee works for, retrieve the employee's name, department number, and salary

```
SELECT E.Fname, E.Lname, E.Dno, E.Salary
FROM EMPLOYEE AS E JOIN WORKS_ON AS W
   ON E.Ssn = W.Essn
WHERE E.Salary > (
   SELECT AVG(E2.Salary)
   FROM EMPLOYEE AS E2
   JOIN WORKS_ON AS W2
     ON E2.Ssn = W2.Essn
   WHERE E2.Dno = E.Dno )
ORDER BY Salary DESC;
```

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

연습 6-2 (2) 오류 찾고 고치기

- 오류를 찾아 고쳐라

✍ 메모

⚠ 주의 오류 ① 바깥 JOIN으로 같은 직원이 중복 출력 ② 서브쿼리 AVG도 조인된 행으로 계산돼 왜곡

✏ 예·풀이 고침: WHERE E.Ssn IN (SELECT W.Essn FROM WORKS_ON AS W) AND E.Salary > (SELECT AVG(E2.Salary) FROM EMPLOYEE AS E2 WHERE E2.Dno = E.Dno) → Wallace 43000, Wong 40000, Narayan 38000



Q&A

- 질문과 답 — SQL (Hands-on Practices)

✍️ 메모

💡 이해 패턴 요약: '가장' → MAX/MIN 서브쿼리 · '모두/만' → NOT EXISTS (... EXCEPT ...) · 0개 그룹도 → LEFT JOIN + 조건은 ON + COUNT(열)